NeuroLog AI
The short guide

What it does: collects application logs and helps us find repeated activity, review possible problems, and understand the evidence.

Remember: Log explorer tells us what happened. Investigations helps us understand it. The AI assistant helps us ask questions.

The main tabs

Tab	Simple explanation
Applications	Choose the application to monitor, such as Guest Connect.
Overview	See recent activity, error counts, and the next investigation step.
Log explorer	Search and filter received logs; inspect the original message.
Investigations	Find repetition, similar messages, unusual messages, and related errors in one place.
AI assistant	Ask about received evidence and get explanations and suggested checks.
Connect applications	Get an ingestion key, endpoint, and example code to send logs.
Settings	Choose appearance, run analysis, and read setup information.

Select an application first. The main analysis views then use that source. A full reload returns to the application picker.

Live app: neurolog-ai.onrender.com

Inside Investigations

View	What it does
Overview	Brings related warnings/errors together and shows incident context.
Named patterns	Finds messages that repeat and gives them recognizable names.
Clusters	Groups messages with similar wording after running clustering.
Outliers	Shows messages isolated by the last clustering run.
Evidence ledger	Filters and exports records; explains risk and supports reversible suppression.
Message space	Shows a visual similarity map and nearest messages.
Alert rules	Defines conditions for attention and shows matching evidence.

Pattern versus incident

A **pattern** means something repeats. An **incident group** collects related warning/error evidence. A normal heartbeat can be a pattern without being a problem. Repeated errors can be both.

Read the evidence, not only the label

Pattern names come from keyword rules and can be misleading. An outlier is unusual wording, not automatically a failure. Risk is a review score, not a probability or confirmed root cause.

Slack, PagerDuty, and webhook alert options currently create **payload previews**; they do not send external notifications.

How we built it and how it works

The technology

Part	Implementation
Interface	React + Vite for pages, navigation, and charts.
Backend	Python Flask for receiving logs and analysis requests.
Storage	SQLite for records, keys, rules, and stored analysis.
Text analysis	TF-IDF converts words into weights; DBSCAN groups similar messages.
Assistant	Local rule-based explanation, with optional Groq language-model responses.
Hosting	Docker packages the dashboard and backend for Render.

A log travels through these steps

Application action -> logger -> delivery queue -> ingestion API -> SQLite -> analysis -> dashboard.

The sender uses a private ingestion key. Queues retry failed delivery. The dashboard refreshes every few seconds using HTTP polling; it is near real time.

What we added

Starting from the original MIT project, we built working ingestion, a Python task source, labeled error generation, analysis tools, guided connections, Guest Connect instrumentation, and application-specific views. We redesigned the UI and combined incidents and patterns into Investigations.

We did not train a new language model. Clustering runs when requested; recurring patterns update automatically. The assistant advises but never executes repairs.

Use it and explain it in a demo

A simple demonstration

1. Select **guest-connect** in Applications.
2. Create or update an event in Guest Connect.
3. Open Log explorer and search **event_created**, **event_updated**, or **save_succeeded**.
4. Show a record ID, source, time, and outcome.
5. Open Investigations to explain repetition and evidence.
6. Ask the assistant to summarize evidence and suggest checks.

Show errors separately

Select task-service and start the optional error scenarios. Generated messages are marked **[GENERATED]**. After at least five logs, run clustering and inspect results. Stop generation afterward.

Why logs appear while idle

Heartbeats, startup, background requests, and page visibility can produce real logs. They are not automatically random generation. Guest Connect records sanitized activity in both demo and live modes; raw typed values and contact details are excluded.

Important limits

Zero errors means none in the recent window, not zero bugs. A click is an attempt; save_succeeded is an outcome. The public demo is login-free, and Render storage is ephemeral: restarts can reset logs and keys. Use non-sensitive development data.

Your closing sentence

“NeuroLog turns application activity into searchable evidence and guided investigation. It helps us decide what to check next.”